

Access your Website and VM

1. To access your webserver from your browser, use your FQDN in your browser;

[for this example, [yourRCSid]RPI will be replaced with rplotka]

[yourRCSid]RPI.eastus.cloudapp.azure.com

youRCSidRPI.eastus.cloudapp.azure.com



The screenshot shows a web browser window with the address bar displaying "rplotka.eastus.cloudapp.azure.com". The page title is "Apache2 Default Page" and the Ubuntu logo is prominently displayed. Below the logo, a message states "It works!". The main content area contains a welcome message for the Apache2 server, explaining that the page is used to test the correct operation of the server after installation. It also provides instructions for users who are not familiar with the page, suggesting they contact the site's administrator. A section titled "Configuration Overview" follows, detailing the default configuration files and their locations. A code block shows the contents of the /etc/apache2/apache2.conf file, which includes references to ports.conf, mods-enabled, conf-enabled, and sites-enabled directories. A list of bullet points explains the roles of these files and how they are managed. Finally, a section titled "Document Roots" is partially visible at the bottom of the page.

← ↻ ⚠ Not Secure | rplotka.eastus.cloudapp.azure.com

Apache2 Default Page

Ubuntu

It works!

This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should **replace this file** (located at `/var/www/html/index.html`) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.

Configuration Overview

Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is **fully documented in `/usr/share/doc/apache2/README.Debian.gz`**. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the **manual** if the `apache2-doc` package was installed on this server.

The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:

```
/etc/apache2/
|-- apache2.conf
|   |-- ports.conf
|-- mods-enabled
|   |-- *.load
|   |-- *.conf
|-- conf-enabled
|   |-- *.conf
|-- sites-enabled
|   |-- *.conf
```

- `apache2.conf` is the main configuration file. It puts the pieces together by including all remaining configuration files when starting up the web server.
- `ports.conf` is always included from the main configuration file. It is used to determine the listening ports for incoming connections, and this file can be customized anytime.
- Configuration files in the `mods-enabled/`, `conf-enabled/` and `sites-enabled/` directories contain particular configuration snippets which manage modules, global configuration fragments, or virtual host configurations, respectively.
- They are activated by symlinking available configuration files from their respective `*-available/` counterparts. These should be managed by using our helpers `a2enmod`, `a2dismod`, `a2ensite`, `a2dissite`, and `a2enconf`, `a2disconf`. See their respective man pages for detailed information.
- The binary is called `apache2` and is managed using `systemd`, so to start/stop the service use `systemctl start apache2` and `systemctl stop apache2`, and use `systemctl status apache2` and `journalctl -u apache2` to check status. `system` and `apache2ctl` can also be used for service management if desired. **Calling `/usr/bin/apache2` directly will not work** with the default configuration.

Document Roots

By default, Ubuntu does not allow access through the web browser to *any* file outside of those located in

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2. Check your PHP installation, using out browser, go to

[http://\[yourRCSid\]RPI.eastus.cloudapp.azure.com/checkphp.php](http://[yourRCSid]RPI.eastus.cloudapp.azure.com/checkphp.php)

← ↻ ⚠ Not Secure | rplotka.eastus.cloudapp.azure.com/checkphp.php

PHP Version 8.1.2



System	Linux rplotka 5.15.0-1017-azure #20-Ubuntu SMP Fri Aug 5 12:00:24 UTC 2022 x86_64
Build Date	Aug 15 2022 12:24:10
Build System	Linux
Server API	Apache 2.0 Handler
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc/php/8.1/apache2
Loaded Configuration File	/etc/php/8.1/apache2/php.ini
Scan this dir for additional .ini files	/etc/php/8.1/apache2/conf.d
Additional .ini files parsed	/etc/php/8.1/apache2/conf.d/10-mysqld.ini, /etc/php/8.1/apache2/conf.d/10-opcache.ini, /etc/php/8.1/apache2/conf.d/10-pdo.ini, /etc/php/8.1/apache2/conf.d/15-xml.ini, /etc/php/8.1/apache2/conf.d/20-calendar.ini, /etc/php/8.1/apache2/conf.d/20-ctype.ini, /etc/php/8.1/apache2/conf.d/20-curl.ini, /etc/php/8.1/apache2/conf.d/20-dom.ini, /etc/php/8.1/apache2/conf.d/20-exif.ini, /etc/php/8.1/apache2/conf.d/20-ffi.ini, /etc/php/8.1/apache2/conf.d/20-fileinfo.ini, /etc/php/8.1/apache2/conf.d/20-ftp.ini, /etc/php/8.1/apache2/conf.d/20-gettext.ini, /etc/php/8.1/apache2/conf.d/20-iconv.ini, /etc/php/8.1/apache2/conf.d/20-mbstring.ini, /etc/php/8.1/apache2/conf.d/20-mysqli.ini, /etc/php/8.1/apache2/conf.d/20-pdo_mysql.ini, /etc/php/8.1/apache2/conf.d/20-phar.ini, /etc/php/8.1/apache2/conf.d/20-posix.ini, /etc/php/8.1/apache2/conf.d/20-readline.ini, /etc/php/8.1/apache2/conf.d/20-shmop.ini, /etc/php/8.1/apache2/conf.d/20-simplexml.ini, /etc/php/8.1/apache2/conf.d/20-sockets.ini, /etc/php/8.1/apache2/conf.d/20-sysvmsg.ini, /etc/php/8.1/apache2/conf.d/20-sysvsem.ini, /etc/php/8.1/apache2/conf.d/20-sysvshm.ini, /etc/php/8.1/apache2/conf.d/20-tokenizer.ini, /etc/php/8.1/apache2/conf.d/20-xmldr.ini, /etc/php/8.1/apache2/conf.d/20-xsl.ini, /etc/php/8.1/apache2/conf.d/20-zip.ini
PHP API	20210902
PHP Extension	20210902
Zend Extension	420210902
Zend Extension Build	API420210902.NTS
PHP Extension Build	API20210902.NTS
Debug Build	no
Thread Safety	disabled
Zend Signal Handling	enabled
Zend Memory Manager	enabled
Zend Multibyte Support	provided by mbstring
IPv6 Support	enabled
DTrace Support	available, disabled
Registered PHP Streams	https, ftps, compress.zlib, php, file, glob, data, http, ftp, phar, zip
Registered Stream Socket Transports	tcp, udp, unix, udg, ssl, tls, tlsv1.0, tlsv1.1, tlsv1.2, tlsv1.3
Registered Stream Filters	zlib.*, string.rot13, string.toupper, string.tolower, convert.*, consumed, dechunk, convert.iconv.*

This program makes use of the Zend Scripting Language Engine:
Zend Engine v4.1.2, Copyright (c) Zend Technologies
with Zend OPcache v8.1.2, Copyright (c), by Zend Technologies



Configuration

apache2handler

Apache Version	Apache/2.4.52 (Ubuntu)
Apache API Version	20120211
Server Administrator	webmaster@localhost

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Access your Website and VM

3. To Access your VM from your computer (without the browser);

From your computer

Open Terminal

Type SSH RSCid@[your FQDN]

```
ssh rplotka@rplotkaRPI.eastus.cloudapp.azure.com
```

```
~ ➤ ssh rplotka.eastus.cloudapp.azure.com
The authenticity of host 'rplotka.eastus.cloudapp.azure.com (20.228.209.240)' can't be established.
ED25519 key fingerprint is SHA256:nkAUtEM/eh/XK3eK0Ar0B5lsVjoaRptQb8mf7TGD1Ss.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

```
ssh rplotka.eastus.cloudapp.azure.com
~ ➤ ssh rplotka.eastus.cloudapp.azure.com
The authenticity of host 'rplotka.eastus.cloudapp.azure.com (20.228.209.240)' can't be established.
ED25519 key fingerprint is SHA256:nkAUtEM/eh/XK3eK0Ar0B5lsVjoaRptQb8mf7TGD1Ss.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'rplotka.eastus.cloudapp.azure.com' (ED25519) to the list of known hosts.
rplotka@rplotka.eastus.cloudapp.azure.com's password: 
```

Access your Website and VM

```
~$ ssh rplotka.eastus.cloudapp.azure.com
The authenticity of host 'rplotka.eastus.cloudapp.azure.com (20.228.209.240)' can't be established.
ED25519 key fingerprint is SHA256:nkAUtEM/eh/XK3eK0Ar0B51sVj0aRptQb8mf7TGD1Ss.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'rplotka.eastus.cloudapp.azure.com' (ED25519) to the list of known hosts.
rplotka@rplotka.eastus.cloudapp.azure.com's password:
Welcome to Ubuntu 22.04.1 LTS (GNU/Linux 5.15.0-1017-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

8 updates can be applied immediately.
1 of these updates is a standard security update.
To see these additional updates run: apt list --upgradable

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

rplotka@rplotka:~$
```

4. Now we are logged on to the other computer. We need to create a location to hold our personal website

First, we need to make sure the web server user www-data has the correct authority to the folders, type:

```
$ sudo chown -R www-data:www-data /var/www/html
```

Web pages are served from /var/www/html, so let's go there:

Type:

```
cd /var/www/html
```

```
ls -la
```

Now we need to create some folders for our website

Create our class root folder;

```
sudo mkdir iit
```

```
sudo chown -R www-data:www-data iit
```

Create our security key folder (more soon)

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```
sudo mkdir /var/www/.ssh
```

```
sudo chown -R www-data:www-data /var/www/.ssh
```

Note: sudo (Super User Do) is a utility that allows us to run commands on our Linux machine with supervisor authority. Just like running an app with supervisor or administrative authority on our computers.

Now let's add ourselves into the www-data group

```
sudo usermod -a -G www-data [yourRCSid]
```

Log out and then log back in

Make sure you are in the group by typing

```
groups
```

You should see www-data in your list of groups

Now let's move into our iit folder

Type:

```
cd /var/www/html/iit
```

Now let's create a simple webpage

Type:

```
sudo nano index.html
```

Type the following in the editor (or cut/paste)

```
<html>
  <head>
  </head>
  <body>
    <p>Hello World!</p>
  </body>
</html>
```

Then type (NOTE: <ctrl> is the control key, and <Enter> is the enter key)

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<ctrl>O followed by <Enter>

Which saves our changes. Then type

<ctrl>X

Which exits and brings us back to the command line

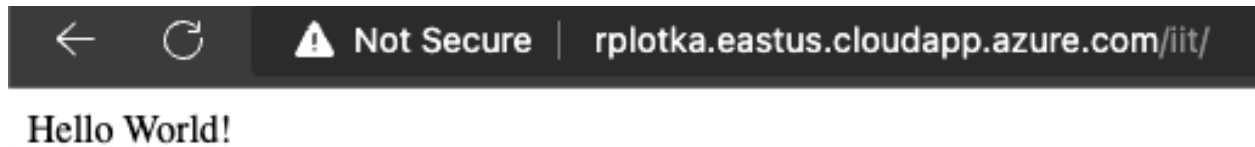
This will put you back at the command prompt.

5. Type ls

You should see index.html listed

Now let's test it by going to FQDN/iit in our browser

Eg



[Print out this page](#)

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6. Now let's remove public access from our sites

First, we need to allow our server to allow authentication

```
sudo nano /etc/apache2/apache2.conf
```

<ctrl-w> and search for `'/var/www/'`

change

AllowOverride None

to

AllowOverride All

```
GNU nano 6.2 /etc/apache2/apache2.conf

<Directory /var/www/>
    Options Indexes FollowSymLinks
    AllowOverride None
    Require all granted
</Directory>

#<Directory /srv/>
#     Options Indexes FollowSymLinks
#     AllowOverride None
#     Require all granted
#</Directory>

# AccessFileName: The name of the file to look for in each directory
# for additional configuration directives. See also the AllowOverride
# directive.
#
AccessFileName .htaccess

#
# The following lines prevent .htaccess and .htpasswd files from being
# viewed by Web clients.
#
<FilesMatch "^\.ht">
    Require all denied
</FilesMatch>

#
# The following directives define some format nicknames for use with
# a CustomLog directive.
#
# These deviate from the Common Log Format definitions in that they use %0
# (the actual bytes sent including headers) instead of %b (the size of the
# requested file), because the latter makes it impossible to detect partial
# requests.
#
# Note that the use of %{X-Forwarded-For}i instead of %h is not recommended.
# Use mod_remoteip instead.
#
LogFormat "%v:%p %h %l %u %t \"%r\" %s %0 \"%{Referer}i\" \"%{User-Agent}i\"" vhost_combined

␣ Help      ␣ Write Out  ␣ Where Is   ␣ Cut        ␣ Execute    ␣ Location   ␣ Undo       ␣ Set Mark   ␣ To Bracket ␣ Previous
␣ Exit      ␣ Read File  ␣ Replace    ␣ Paste       ␣ Justify    ␣ Go To Line ␣ Redo       ␣ Copy       ␣ Where Was  ␣ Next
```

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```
GNU nano 6.2 /etc/apache2/apache2.conf *
<Directory /var/www/>
    Options Indexes FollowSymLinks
    AllowOverride All
    Require all granted
</Directory>

<Directory /srv/>
    # Options Indexes FollowSymLinks
    # AllowOverride None
    # Require all granted
</Directory>

# AccessFileName: The name of the file to look for in each directory
# for additional configuration directives. See also the AllowOverride
# directive.
#
AccessFileName .htaccess

#
# The following lines prevent .htaccess and .htpasswd files from being
# viewed by Web clients.
#
<FilesMatch "\.ht$">
    Require all denied
</FilesMatch>

#
# The following directives define some format nicknames for use with
# a CustomLog directive.
#
# These deviate from the Common Log Format definitions in that they use %O
# (the actual bytes sent including headers) instead of %b (the size of the
# requested file), because the latter makes it impossible to detect partial
# requests.
#
# Note that the use of %[X-Forwarded-For] instead of %h is not recommended.
# Use mod_remoteip instead.
#
LogFormat "%v:%p %h %l %u %t \"%r\" %s %O \"%{Referer}i\" \"%{User-Agent}i\"" vhost_combined
```

7. Now, let's create a file to allow our iit folder to be seen by authenticated users

```
sudo nano /var/www/html/iit/.htaccess
```

cut/paste the following

AuthType Basic

AuthName "Restricted Content"

AuthUserFile /etc/apache2/.htpasswd

Require valid-user

```
GNU nano 6.2
AuthType Basic
AuthName "Restricted Content"
AuthUserFile /etc/apache2/.htpasswd
Require valid-user
```

Now let's add some users and passwords

For the first user – you – we need the -c flag

```
sudo htpasswd -c /etc/apache2/.htpasswd [yourRCSid]
```

enter a password

NOTE – omit the -c after the first time you type the command

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```
sudo htpasswd /etc/apache2/.htpasswd rplotka  
sudo htpasswd /etc/apache2/.htpasswd [TA RCSid][mojiso]
```

```
rplotka@rplotka:/var/www/html$ sudo htpasswd /etc/apache2/.htpasswd rplotka  
New password:  
Re-type new password:  
Adding password for user rplotka  
rplotka@rplotka:/var/www/html$ sudo htpasswd /etc/apache2/.htpasswd achara  
New password:  
Re-type new password:  
Adding password for user achara
```

let's check by typing
cat /etc/apache2/.htpasswd

```
rplotka@rplotka:/var/www/html$ cat /etc/apache2/.htpasswd  
rplotka:$apr1$zVWcRIOB$FQJW.XoF59n89BJvH6XYW/  
achara:$apr1$YXCyDQ51$v4EhaEjutid5NklvEj3xP1  
rplotka@rplotka:/var/www/html$
```

Let's restart our server to have our changes take effect
sudo service apache2 restart